# What Counts as a Mistake? Annotating Recitation Events in Quran Memorization Transcripts

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## Abstract

Checking Quran recitation from an ASR transcript requires distinguishing unresolved mistakes from repetitions, repairs, opening formulas and accepted spelling differences. We report a completed human annotation of 100 production recording cases, covering 348 scored units and 162 localized events across ten combined labels. An executable evaluator scores labels and word positions together. A plain diff reaches label-aware F1 0.531 and localization F1 0.839; adapted production cleaner/alignment components reach 0.518 and 0.799. Their exact-span F1 scores are both 0.505. Correcting the adapter's word coordinates recovers all five annotated repetition events, showing why annotation interfaces must be checked before interpreting baseline failures. The dataset and scorecard support a planned comparison between agents developing annotation methods under a shared budget.

## 1 The distinction the task turns on

Quran memorization is normally practised with a listener who corrects the reciter. An application that stands in for that listener transcribes with commercial ASR and then compares the transcript against the reference. Edit distance locates differences but does not determine their interpretation. In our production data 74% of reviewed ayah units carry no event at all, and among those that do, a substantial share are things a human listener would never mention: the reciter repeated a phrase after a breath, said a word wrongly and immediately fixed it, or used a written form differing from the Uthmani orthography without differing in what was recited. Flagging these as unresolved mistakes can give the learner misleading feedback; we do not measure the relative user cost here.

We therefore annotate transcript-against-reference differences as typed, localized *events*, and we judge the text rather than the speaker: a label records what differs between transcript and reference, not whether the reciter or the ASR caused it. Unwritten vowel and tajweed errors lie outside a transcript-only rubric and outside this work.

## 2 Taxonomy and annotation

**Combined labels.** An earlier pilot used two parallel fields, an event type and a benign/mistake verdict, plus an `uncertain` escape hatch. Reviewing real recordings made both untenable: the verdict was never independent of the type in practice, and `uncertain` absorbed exactly the cases that needed a decision. The rubric now uses one combined label per event, `substitution_mistake`, `omission_mistake`, `insertion_mistake`, `substitution_corrected`, `omission_corrected`, `repetition_benign`, `letters_benign`, `spelling_benign`, `basmala_benign` and `isti3adha_benign`, with `clean` for a reviewed unit carrying no event. A unit reviewed and found clean is not the same as a unit not yet reviewed, and the data distinguishes them.

**Four decisions that took adjudication.** *Omissions* include missing beginnings, interiors and endings, with no exemption for the final ayah of a recording: stopping early still leaves reference material absent from the transcript, and a reviewer cannot infer intent from recording position. *Repeats* span the original phrase together with its extra occurrence against a single reference copy, aligning their localization with corrected events; a wrong-then-correct attempt is a corrected event, while a correct-then-incorrect restatement is a `substitution_mistake` spanning both attempts, with the order recorded in the note. *Spelling* differences removed by an agreed context-independent normalization receive no event, and only a residual form needing contextual acceptance receives `spelling_benign`; the accepted contexts are enumerated rather than generalized, so the category cannot quietly become blanket final-letter deletion. *Localization* uses verbatim words and half-open spans into the original whitespace token arrays, with an empty hypothesis span anchoring an omission and an empty reference span anchoring an insertion.

**Scale and composition.** All 100 selected recording cases are human-approved: 314 ayah units, 233 of them clean, 115 within-ayah events and 47 opening-formula annotations, over 274 ayahs in 38 surahs. The event distribution is uneven by nature, from 62 `substitution_mistake` down to one `insertion_mistake` and one `omission_corrected`, and any evaluation has to survive that. Ten learner identities contribute, one of them 46 recordings. Selection was purposive, so these counts describe the annotated set; they do not estimate error prevalence or imply generalization to unseen reciters. Three further approved recordings remain outside the scored set; their transcript-only copies join the development candidates, with annotations archived privately. We verified identifiers and transcripts against both source exports and excluded punctuation-normalized duplicates and redundant clean subpassages. Gold answers and selected membership remain in an owner-held private bundle; this draft does not release a gold sample.

## 3 An executable scorecard

Systems receive the reviewed ayah split, the ayah ids and the exact references, so detection and segmentation are supplied rather than scored; that task is treated in a companion submission. Per unit a system returns events of the form `{label, hyp_span, ref_span}`, with half-open spans into the transcript and reference token arrays.

**Opening formulas are scored.** Each record's isti'adhah and basmala are presented as their own unit, with the formula as the transcript and an empty reference; a compound opening yields two events over one token array. Deciding that a basmala preceding the ayah is benign rather than an insertion is exactly the judgment under test, and excluding it would hide the failure mode that turns out to matter most.

**Matching.** Predictions and gold events are paired one-to-one within a unit by maximum total similarity, so a single broad prediction is credited with at most one gold event and the remainder count as misses. A pair is eligible only when *both* spans overlap, $\min(\mathrm{sim}_{hyp}, \mathrm{sim}_{ref}) \geq 0.30$; averaging the two, as an earlier version did, lets an exact reference span carry a match while the hypothesis span points somewhere else entirely. Events are validated for known labels, integer endpoints, token bounds and label-specific empty spans. Invalid predictions count as false positives. Empty spans match only empty anchors, exactly or within one token; they never match selected words. Pairing ignores labels so localization can be reported separately. Larger units with more than seven events on either side use deterministic greedy one-to-one matching.

**Scores.** The primary measure is label-aware event F1: a paired prediction is a true positive only when its label also matches, and a mislabeled pair counts once as a false positive and once as a false negative. We report micro F1 as the headline, exact-span F1 requiring identical endpoints, macro F1 across labels present in gold, and localization precision/recall/F1 with labels dropped. An application-facing cost is reported separately in raw event counts, $(2\,\text{missed} + \text{false flags})$ per 100 units, so that the stated exchange rate of one missed mistake for two unnecessary flags is the one the formula implements. The 2:1 ratio is a stated preference, not an empirically established user cost. Predicting nothing gives F1 0 and the gold annotation gives 1. Evaluator v2.1 passes synthetic coordinate, label, anchor and adapter regression tests and private gold oracle checks. The tolerance of

0.30, one-token anchor slack and secondary cost remain provisional until disjoint practice review; baseline results are conditional on these choices.

## 4 What existing systems do

| System | micro F1 | exact F1 | macro F1 | loc F1 | review cost |
|---|---|---|---|---|---|
| Predict nothing | 0.000 | 0.000 | 0.000 | 0.000 | 51.2 |
| Plain diff | 0.531 | 0.505 | 0.179 | 0.839 | 22.4 |
| Adapted production components | 0.518 | 0.505 | 0.284 | 0.799 | 26.7 |

Table 1: Evaluator v2.1: 348 units, 162 gold events. All baseline calls complete without crashes or invalid predictions.

The plain diff locates 128 of 162 gold events (recall 0.790), with localization F1 0.839. Its label-aware recall is 0.500 and F1 is 0.531. These quantities differ: localization F1 is not the percentage of gold events found. It reaches per-label F1 0.94 for omissions and 0.85 for substitutions, and zero for the other eight labels. It emits 46 insertion predictions, 34 of them on opening-text units, whereas gold has one insertion event. Compound openings contain two gold formulas but a diff can return one insertion spanning both.

The second baseline adapts existing production cleaner and alignment components to the new event schema; the deployed application does not natively expose it. The adapter retains original token identities through deletion, uses alignment indices and reference offsets, and maps cleaner notes to reference phrases and both attempts. Its explicit heuristic interprets a stumble as a repair only when the following equally long surviving span matches the reference; unlocalizable removed fragments remain extra text. Production's partial-start scoring policy is retained. These choices are part of the baseline, not additional gold decisions.

This adapter predicts five repetitions and recovers all five (F1 1.0); six corrected-substitution predictions recover one of four gold events (F1 0.20). It has nonzero F1 on four labels and zero on six, and never predicts either opening-formula label, letters, contextual spelling or omission corrections. The earlier adapter lost valid repetition credit through incorrect coordinates. The remaining label gaps motivate explicit event modeling, but these results do not establish that cleaning itself causes the errors or measure every feature of the deployed application. Each baseline retains its own comparison normalization, which is an algorithm choice rather than a change to the gold rubric.

## 5 Limitations and what comes next

The annotated set is purposive and small, one learner contributes 46 of the 100 recordings, and the label distribution is heavily skewed, so macro F1 here is a coverage check rather than a stable estimate. More consequentially, all 100 selected transcripts already appear as inputs in a public release of the same recordings, and twelve appear in an earlier machine-labeled pilot. The adjudications and the selection membership are private, but the inputs are not, and runtime isolation cannot establish absence of prior exposure. An unseen-input claim requires a separate corpus. The prepared development release contains 127 cases and 888 units, including 44 of 63 newer recordings. It permits shared clean ayahs and different error variants, but excludes complete gold copies and copies of event-bearing gold chunks. The one local ASR repair appears non-Quranic and is excluded. Remaining new recordings are not a frozen hidden test.

We plan to compare agents under the same unlabelled training data, rubric, examples, tools and budget. Agents may annotate, train models or write rules; annotation is an available strategy, not the experimental contrast. We will preserve code, logs and generated labels, then score frozen methods privately. Runs have not occurred; agent versions, repeated runs, budgets and feedback remain to be fixed. Any annotation-delivery requirement must apply equally to all agents. A high score supports the combined process, not each training label: code may ignore or correct its labels. Generated annotations need independent review before being treated as reliable; a fully human-labelled training pool is unnecessary. Gold answers, selected test bundles and review history stay outside development environments, with shared clean inputs and prior public exposure disclosed.